

Deliverable 3

Project Title: Process Modeling for "FindMyMeal"

Class Details: IS436 Structured Systems Analysis
and Design

Date: 10/27/2024

Team Name: FindMyMeal

Team Information for FindMyMeal Project

- **Team Name:** FindMyMeal
- **Project Contact Email:** hunnaia1@umbc.edu

Team Members, Roles, and Contact Information

1. **Hunnain Arsalan - Team Lead / Frontend Developer**
 - **Bio:** Leads the team in design and development of the frontend, ensuring the user interface is intuitive and aligns with project requirements.
2. **Muhammad Wajahat - Project Manager**
 - **Bio:** Responsible for overseeing the project's progress, ensuring timely completion of deliverables, and coordinating with team members to meet project goals.
3. **Benaya Berhane - Database Developer**
 - **Bio:** Specializes in database management, focusing on structuring and optimizing data storage to support the recipe and user information needs of the application.
4. **Bazen Mekonen - Backend Developer**
 - **Bio:** Develops and maintains the backend services, ensuring smooth integration with databases and APIs for recipe retrieval and account management.
5. **Ishan Giri - Networking Specialist**
 - **Bio:** Oversees network-related functionalities, ensuring secure and reliable connectivity for data exchanges within the application.
6. **Michael Glass - Database Developer, Testing**
 - **Bio:** Works on database management and testing phases, ensuring data accuracy and testing functionality to meet project standards.

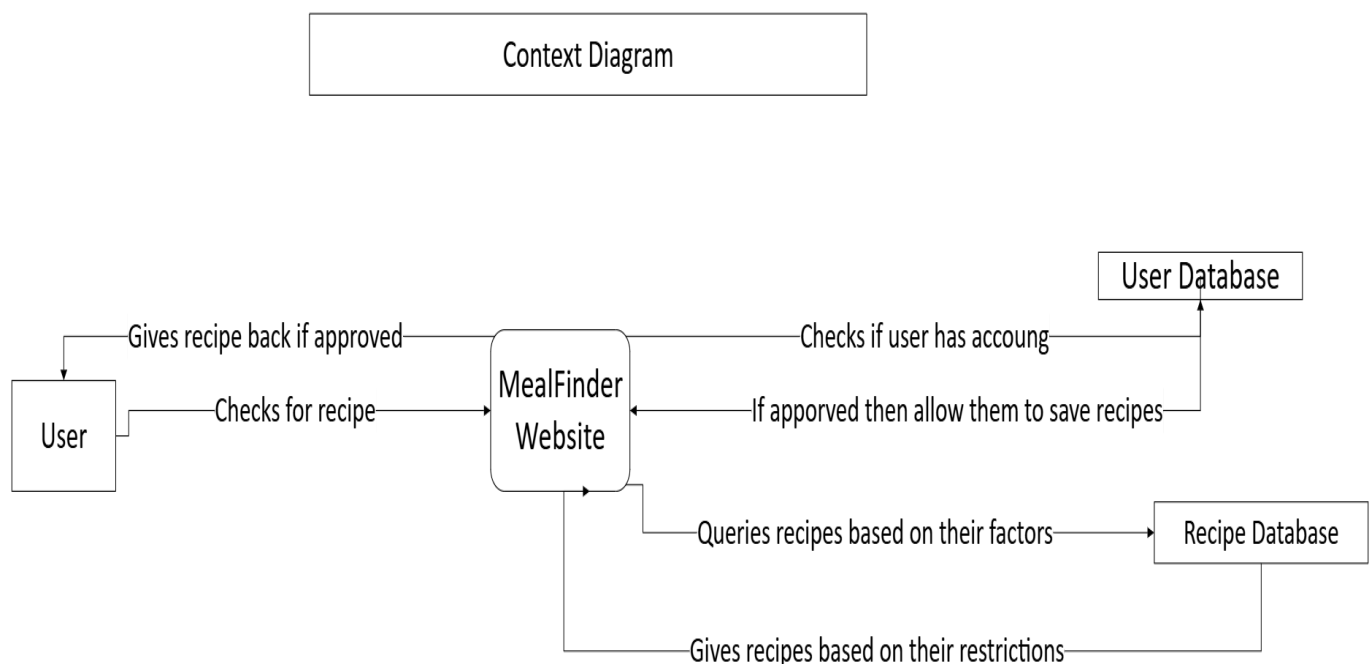
Team Meeting Schedule

Meeting Times: Mondays and Wednesdays at 4 pm

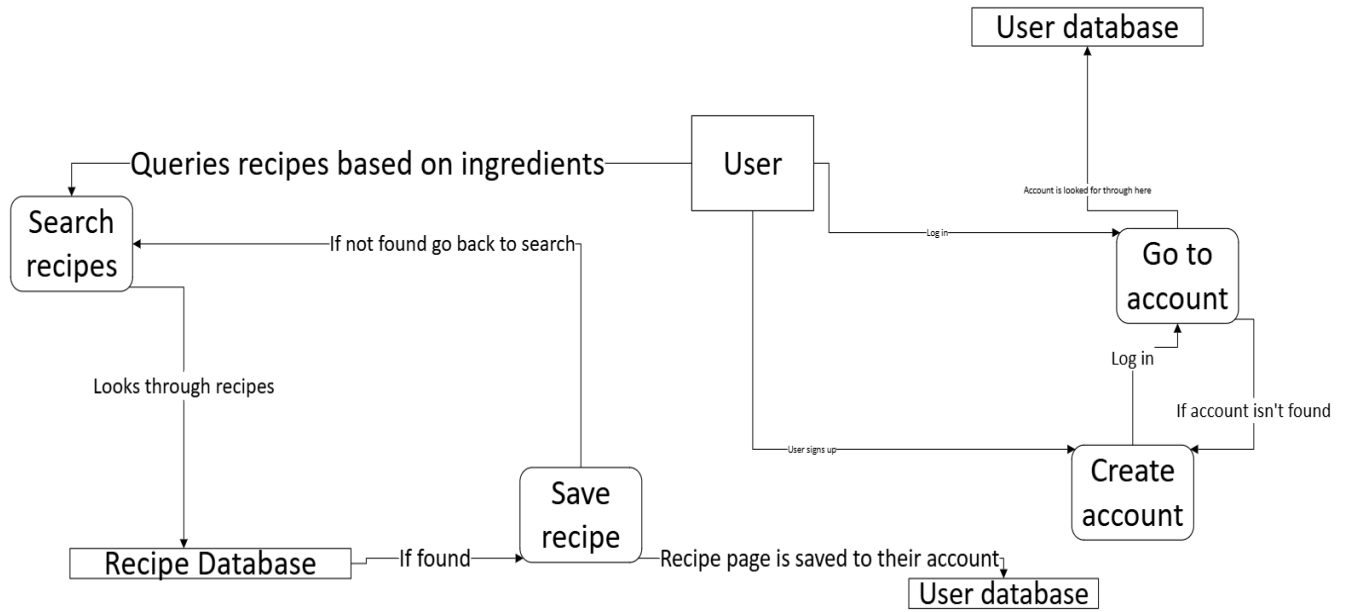
1. Introduction

This document is Deliverable 3 for the "FindMyMeal" project. It focuses on **Process Modeling**, where we utilize Data Flow Diagrams (DFDs) to visualize the system's data movement and refine requirements based on the logical flow of information. This deliverable includes Context, Level 0, and Level 1 DFDs, detailed requirements, updated use cases, and an enhanced project plan.

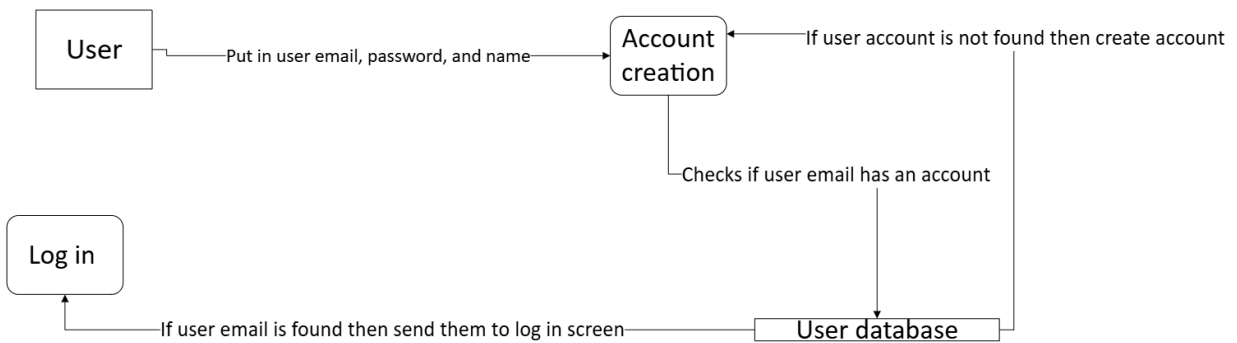
2. Data Flow Diagrams (DFDs)

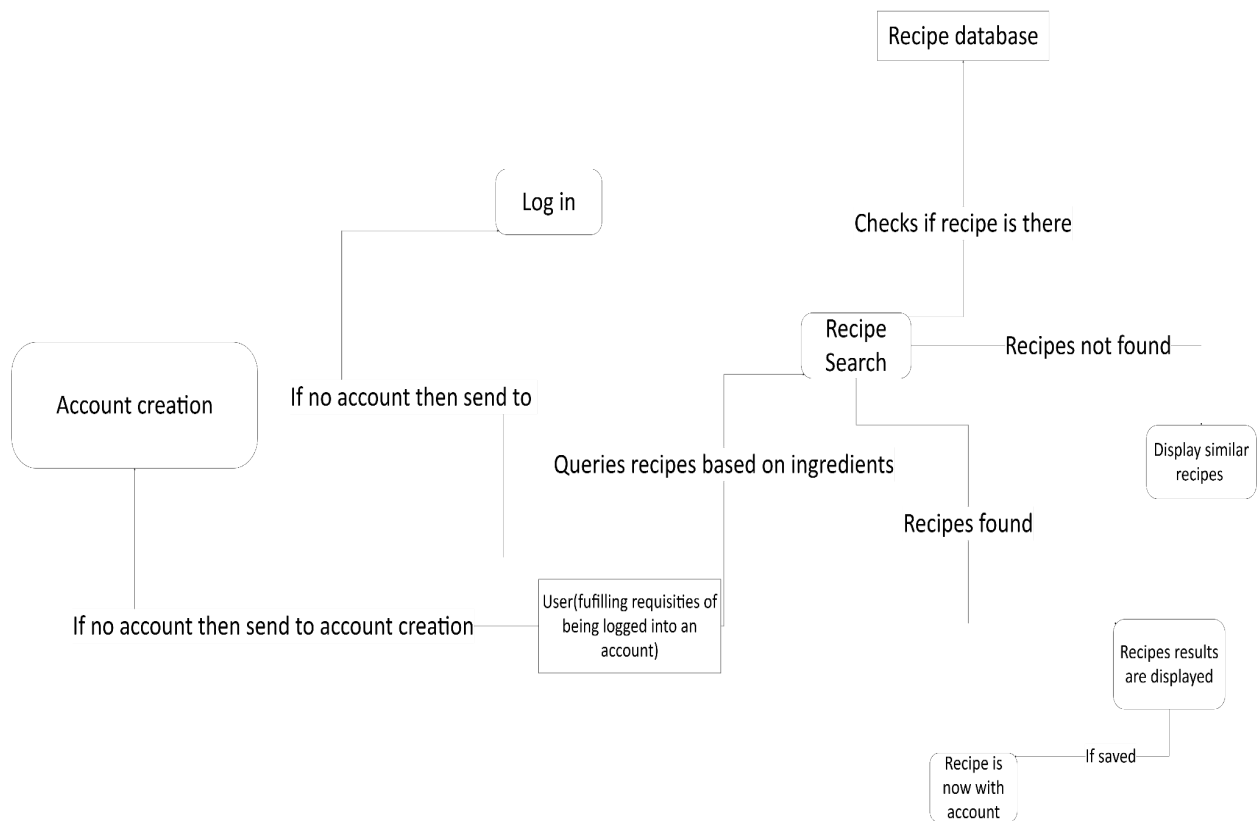
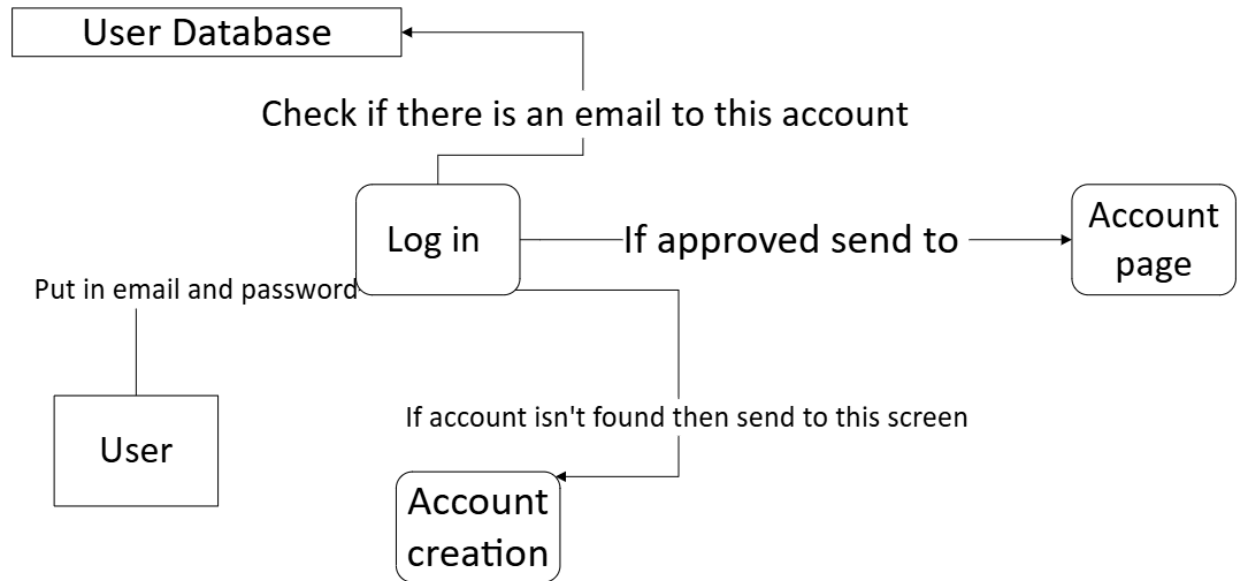


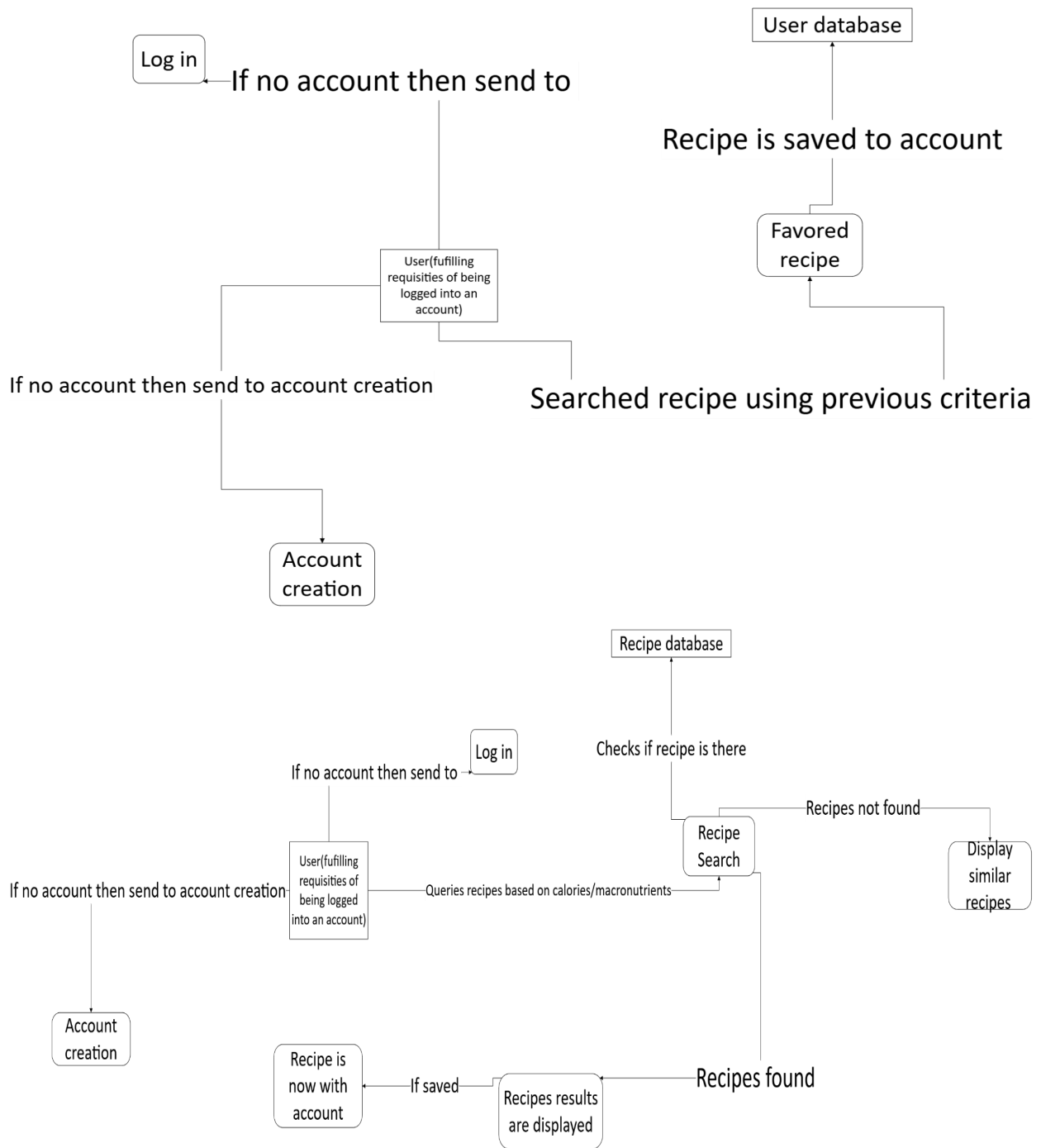
Level 0



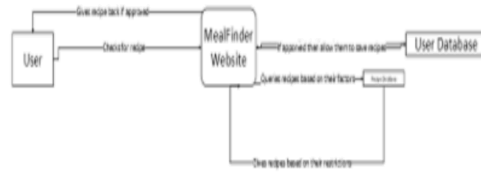
Level 1



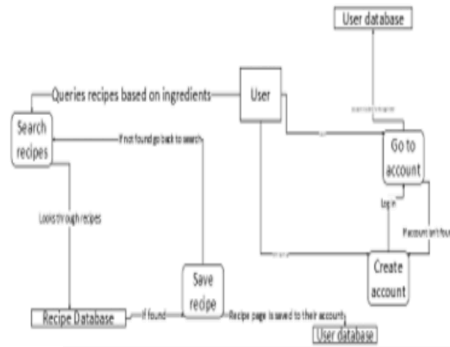




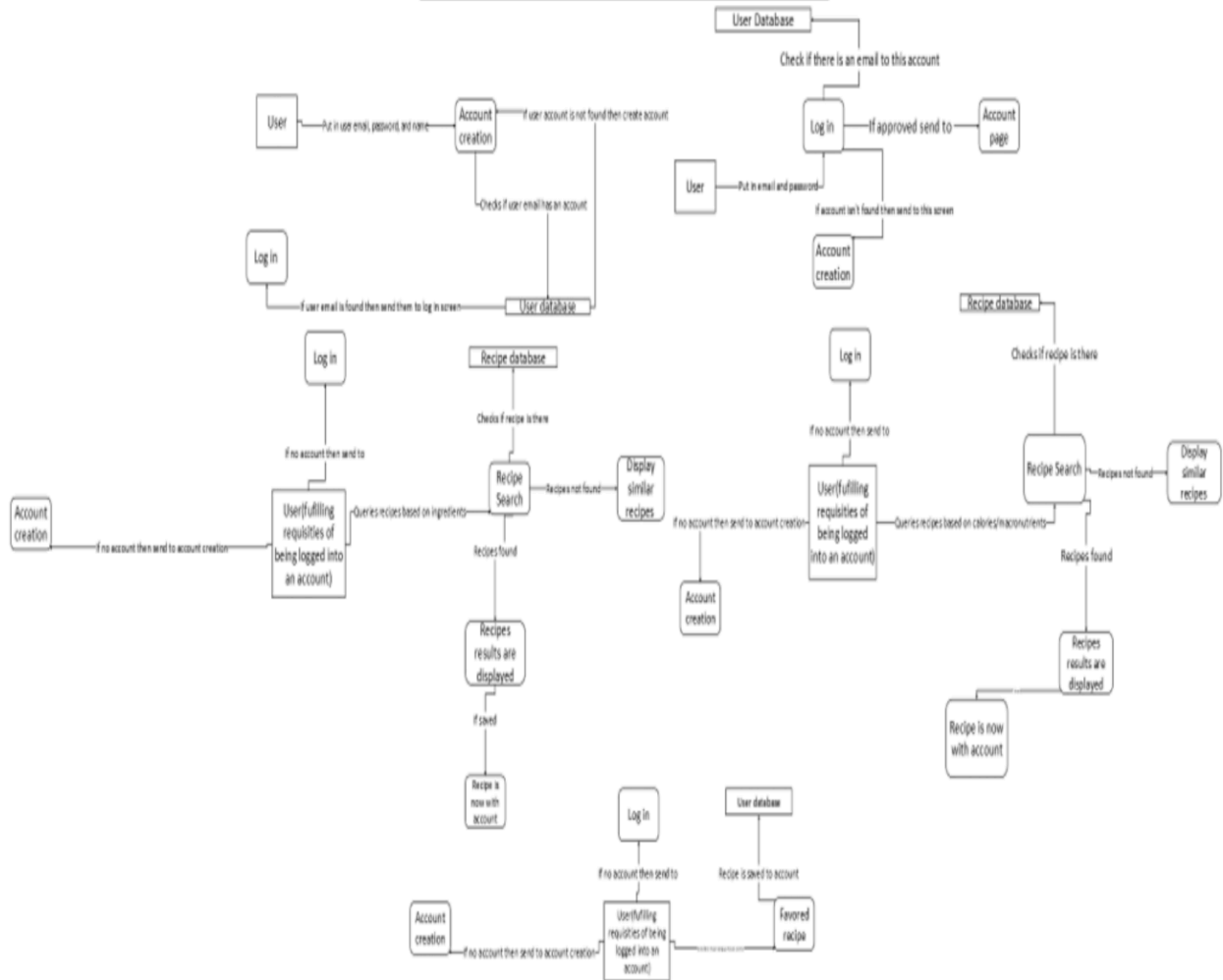
Context Diagram



Level 0



Level 1



2.1 Definitions of processes, data stores and data flows

Processes:

Account Creation

- **Definition:** This process allows users to create an account on the FindMyMeal platform. It collects basic information such as email, password, and username, verifies the validity of the email, and creates a unique profile in the User Database. If the user provided information is valid, a new account is generated and a confirmation message is sent.
- **Data Flows:**

Login:

- **Definition:** This process allows registered users to access their accounts by entering their email and password. The system validates the credentials against stored records in the User Database. If the credentials match, the user is granted access and directed to the main interface. Otherwise, an error message is displayed

Recipe Search

- **Definition:** The Recipe search process enables users to find recipes by entering search criteria, such as specific ingredients or nutritional requirements (eg. macro ratios, calorie limits). The search query is sent to the API, which returns recipe results that match the user's criteria. The user can browse the results, apply additional filters, and view detailed recipe information.

Save Recipe

- **Definition:** This process allows the users to save a selected recipe to their personal saved recipes list in the User Database. When a user selects the Save Recipe option, the system stores the recipe details in their account, allowing easy access for future reference. IF the save action is successful, a confirmation message is displayed.

Unsave Recipe

- **Definition:** The process provides users the option to remove a previously saved recipe from their saved list. When the user clicks the Unsave Recipe option, the system deletes the selected recipe from the user's saved list in the User Database. A confirmation message is shown after successful removal.

Customer Support Communication

- **Definition:** This process handles any user inquiries or issues related to account management or recipe saving functions. Users can contact support for assistance with errors, such as difficulty saving recipes or issues with their account. The support team provides solutions or troubleshooting steps.

2.2 Entities

User

- **Definition:** The primary actor who interacts with the FindMyMeal platform. The User can perform various actions, including creating an account, logging in, searching for recipes based on ingredients or nutrition, saving or unsaving recipes, and accessing customer support.

2.3 Data stores

User Database

- **Definition:** This datastore contains all user related data, including account information (email, password, username) and saved recipes. It plays a crucial role in managing user authentication and personalization of content based on saved preferences.
- **Stored Information:**
 1. User account information: Data for each registered user, including email, password, and username.
 2. Saved Recipes: A list of recipes saved by each user for future reference, which can be modified.

Recipe Database

- **Definition:** A centralized collection of recipe data that the system can access for queries and searches. This database is populated by the Spoonacular API and contains recipe details like ingredients, nutritional information, cuisine type, and preparation time.

- **Stored Information:**

1. Recipe Details: Information on individual recipes, including ingredient list, calories, preparation steps, and servings.
2. Nutritional information: Data on calories and macronutrient distribution, which helps users find recipes that match their dietary needs.

3. Requirements and Use Cases

3.1 Updated Requirements Definition

Functional Requirements:

1. **Account Management:**

- The system must allow users to create an account with unique credentials.
- The system must allow users to log in with valid credentials.

2. **Recipe Search:**

- Users can search for recipes by ingredients or nutritional values.
- Recipe details like calories, servings, and prep time should be displayed.

3. **Recipe Saving:**

- Users can save recipes to their account for easy future access.

Information-Oriented Requirements:

- Saved recipes should be accessible only upon login.
- User preferences and recent searches should personalize future recommendations.

Non-Functional Requirements:

1. **Performance:**

- Recipe retrieval response time should be ≤ 3 seconds.
- The system must support at least 100 users simultaneously.

2. **Security:**

- Data should be encrypted with AES standards.
- Only authorized personnel may access sensitive data.

3.2 Revised Use Cases

Use Case 1: Account Creation

- **Preconditions:** User does not already have an account.
- **Trigger:** User selects the option to sign up.

- **Normal Course:**
 1. User enters email, password, and username.
 2. System checks for existing account.
 3. If none found, a new account is created.
- **Alternative Course:** If email is invalid, user is prompted to re-enter.
- **Post-Conditions:** User account is created and stored.

Use Case 2: Login

- **Preconditions:** User has an account.
- **Trigger:** User enters login credentials.
- **Normal Course:**
 1. User enters email and password.
 2. System checks credentials; if valid, access is granted.
- **Alternative Course:** If credentials are incorrect, display error.
- **Post-Conditions:** User gains account access.

Use Case 3: Finding a Recipe by Ingredients

- **Preconditions:** User is logged in and has specified ingredients.
- **Trigger:** User initiates recipe search with ingredients.
- **Normal Course:**
 1. User enters ingredients.
 2. System queries Recipe Database; displays matches.
- **Alternative Course:** If no match found, suggest similar recipes.
- **Post-Conditions:** Display results matching entered ingredients.

Use Case 4: Finding a Recipe by Calories/Macronutrients

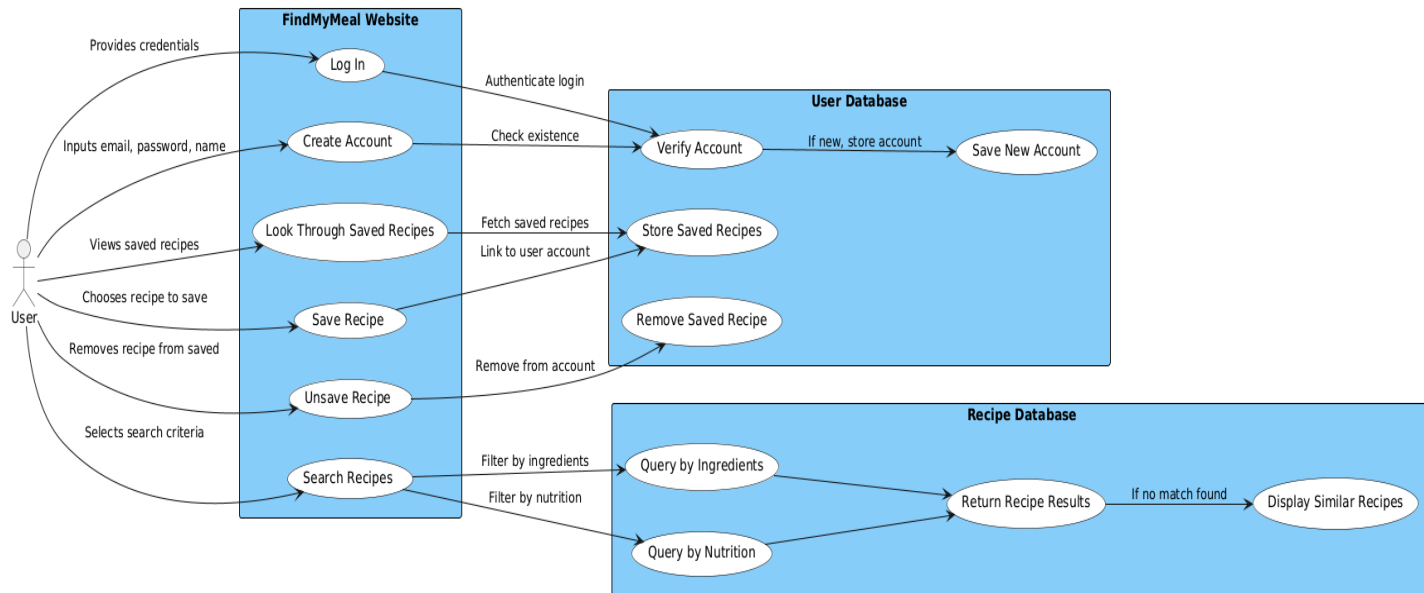
- **Preconditions:** User is logged in with nutritional preferences set.
- **Trigger:** User enters dietary restrictions.
- **Normal Course:**
 1. User enters calorie/macronutrient preferences.
 2. System searches Recipe Database for matches.
- **Alternative Course:** Suggest similar recipes if no match.
- **Post-Conditions:** Recipes that meet preferences are displayed.

Use Case 5: Saving a Recipe

- **Preconditions:** User is logged in and viewing a recipe.

- **Trigger:** User selects "Save Recipe."
- **Normal Course:**
 1. User clicks save; recipe is saved to User Database.
- **Alternative Course:** If error, prompt retry.
- **Post-Conditions:** Recipe is accessible in the user's saved recipes.

3.3 Revised Use Case Diagram



5. Updated Project Plan

Task	Start Date	End Date
Feasibility Analysis	Completed	
Requirements Gathering	10/08/2024	10/14/2024
System Design with DFDs	10/15/2024	10/27/2024
Development Phase	10/29/2024	11/25/2024
Testing Phase	11/26/2024	12/10/2024
Deployment	12/11/2024	12/15/2024
Final Presentation and Report	12/16/2024	12/20/2024

